



INTERNATIONAL UNIVERSITY OF GRAND-BASSAM
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SCHOOL OF SCIENCE AND ENGINEERING

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Requirements II — Interviewing, Observing and Prototyping

How to ask without inventing the answer, how to write down what you saw without deciding what it means, and how to provoke a reaction from someone who cannot describe what they want

Session 8 of 28 · Friday 2 October 2026 · 10:00 – 11:20 AM

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Requirements II — Interviewing, Observing and Prototyping

SESSION 8 OF 28

Friday, 2 October 2026 · 10:00 – 11:20 AM · Unit 2 · Requirements

LEARNING OBJECTIVES

- ✓ Design an interview as a sequence of open, probing and confirming questions — and recognise and repair a leading question.
- ✓ Write an elicitation record that separates what you saw from what you think it means, and states how the inference will be tested.
- ✓ Recognise a workaround as evidence of an unmet requirement, and turn it into a story with a number attached.
- ✓ Run a paper-prototype session that produces behaviour rather than opinions about appearance.

FROM SESSION 7

You have a stakeholder map, an empty chair, and a plan whose order is fixed: documents, then observation, then interviews, then a prototype. Today is the craft of executing it — because each of those three instruments

has a characteristic way of going wrong, and each of those failures produces data that looks fine and is worthless.

TEAM PROJECT · 0

first 15 minutes of class

The SANTE-BASSAM elicitation plan — who does what, on which day

Applying Session 7. The map is drawn; what the team lacks is names and dates against it. Fifteen minutes now saves a fortnight of nobody going anywhere.

WHAT HAPPENS

1. **Reconcile the map.** Each of you added a ninth stakeholder in homework. Put all six on the board, keep the ones that survive challenge, and agree the final grid. **Five minutes.**
2. **Assign the instruments.** Three pairs, one strand each — and every strand names a day and produces a record. **Six minutes.**
3. **Agree the empty chair.** Write one sentence in the log naming who will do backups, upgrades and repairs after hand-over. If the honest answer is “nobody”, write *that*, and it becomes a requirement in Session 9. **Four minutes.**

Pair A — documents

paper register, patient card, Ministry template. Output: the field list, before anyone interviews anybody

Pair B — the desk

two clinic mornings observing the clerk. Output: a three-column record, at least seven rows, with counts

Pair C — outreach

one full day shadowing a CHW. Output: a three-column record, plus battery and connectivity readings taken hourly

Output: the elicitation plan in the progression log — strand, pair, date, what the record will contain. Every technique gets one line saying why it, and not another.

Next session's slot: the three records come back and the team turns the strongest rows into candidate requirements.

1 · Interviewing without inventing the answer

An improvised interview yields anecdotes. A structured one yields requirements. The structure is four question types, used in roughly this order, and one type that destroys the data.

Open — O

STARTS A TERRITORY

Invites a narrative and lets them choose what matters. Anchor it to a concrete recent event, not to an abstraction.

Good: “Walk me through last Tuesday, from when you arrived until the first patient left.” **Weak:** “Describe your workflow.”

Probing — P

GOES AFTER THE EXCEPTION

Follows something they just said, toward the case that breaks the normal path. Most requirements live in the exceptions.

Good: “You said the clerk brings you the card. What happens when she cannot find it? How many times last week?”

Confirming — C

PLAYBACK

You say back what you think you heard, deliberately slightly wrong, and let them correct you. Corrections are worth more than agreements.

Good: “So follow-up is decided at the end of the visit, written in the notebook, checked on Mondays — and the register is never consulted. Right?”

Referral

THE CHEAPEST QUESTION IN THE COURSE

“Who else should we talk to? Who would disagree with you?” This one question found the pharmacy assistant in week 4, and with her a new story and a timing constraint.

Leading — L

DESTROYS THE DATA

Any question that contains its own answer. It proposes a solution, assumes a need, or invites a polite yes — and whatever they reply is worthless, because you cannot tell agreement from courtesy.

Bad: “You’d want to search by phone number, wouldn’t you?” invents a requirement. **Repaired:** “When you need to find an existing patient, what do you use?” discovers one.

WHY THE LEADING QUESTION IS THE WHOLE PROBLEM IN ONE SENTENCE

A leading question is a hypothesis the interviewer is hoping to confirm. Your stakeholder is being polite, possibly deferential, and quite likely wants you to succeed. They will agree. You will write it down as a finding. Nobody in the room will ever know it was yours.

APPLIED TO SANTE-BASSAM

The nurse-in-charge protocol — twelve questions, typed, and two leading first drafts repaired

#	Type	Question	Why it is there
1	O	Walk me through last Tuesday, from when you arrived until the first patient left.	Concrete and recent; opens the whole workflow
2	P	You said the clerk brings you the card. What happens when she cannot find it?	The exception behind S-05
3	P	How many times last week? What did each of those patients do?	Turns an anecdote into a frequency, and a frequency into a priority
4	O	Tell me about the notebook I saw beside the register.	The workaround — the source of S-12
5	P	Pick one name in it. Why is that person there, and when will you take them out?	The tacit rule, extracted one instance at a time
6	P	When a child is brought by a neighbour and the mother is not here, what do you record and what do you refuse?	Consent and identity — a domain requirement in disguise
7	O	What does the district officer send back to you, and why?	The report seen from the other end; detects the conflict
8	P	The last time the power went off during a consultation — what did you do with the record?	Data integrity in her words, with a real frequency attached
9	C	So: follow-up is decided at the end of the visit, written in the notebook, checked on Mondays — and the register is never consulted. Right?	Playback; invites correction
10	O	If this system did one thing wrong, what would make you stop using it the same day?	The non-negotiable — feeds straight into the Definition of Done
11	C	Who else should we talk to? Who would disagree with you?	Referral. This is the question that found the pharmacy assistant

12

C

May we come and sit at the desk on Thursday morning?

Sets up the observation — ask at the end, when trust exists

Leading · first draft

“You’d want the follow-up list on the tablet too, wouldn’t you?”

Proposes a solution (tablet), assumes a need (follow-up while moving) and invites a polite yes. Three faults in eleven words.

Repaired → Q4 / Q5

“Tell me about the notebook.” Then: “Where are you when you look at it?”

If the answer is *only at the desk, on Monday*, then the tablet version was never a requirement — and you have saved a sprint.

Leading · first draft

“Losing a record in a power cut would be unacceptable, right?”

Nobody says no. It also skips the interesting question, which is what she does about it today.

Repaired → Q8

“The last time the power went off — what did you do with the record?”

“I write it on paper and copy it later, and twice I forgot.” That single answer gives the requirement, its rationale *and* a frequency.

2 · Separating what you saw from what you think it means

The most common defect in a student elicitation record is the silent leap from observation to conclusion. You watch the clerk check a number twice, you write down “the system needs validation”, and the thing you actually saw is gone forever. The fix is mechanical: three columns, always.

THE THREE-COLUMN RECORD

Observation

What a camera would have recorded. No verbs of interpretation — no *because*, no *needs*, no *struggles with*. Counts wherever you can get them.

Inference

What you think it means. This is allowed to be wrong; that is what the third column is for.

How to test it

The question you will ask, or the thing you will count, to find out whether the inference holds. An untested inference is not evidence.

APPLIED TO SANTE-BASSAM

Seven rows from one clinic morning at the front desk

#	Observation — what I saw	Inference — what I think it means	How I will test it
1	Clerk copied the patient number from register to card, then checked it twice.	Transcription errors are frequent enough to be worth double-checking.	Ask about a past error. Count discrepancies across 50 cards.
2	Three patients turned away: the register book was with the supervisor.	A single paper copy creates availability failures.	Ask how often. Check whether a second book exists.
3 ★	The nurse keeps a private notebook of patients to follow up.	The official register does not support the follow-up workflow at all.	Ask what is in it and why it is not in the register. (<i>Done</i> — Q4, Q5.)
4 ★	Of 31 returning patients, 19 gave the child's name and their own; the clerk found 17 by scanning for the mother; 2 were registered again.	The mother's name is the primary lookup key in practice. Duplicates run at about 6% of returning visits.	Count duplicate-looking entries in 200 register rows.
5	Clerk wrote "Modeste", "Modest" and "Modesté" for the same village in one morning.	Free-text village names are why the officer rejects reports.	Ask the officer how many reports she returned last quarter, and why.
6	Median time to register a new patient on paper: 75 s over 9 registrations.	Anything slower than this will be abandoned, whatever we build.	Time the same task on the prototype with the same clerk.
7	Power failed 11:20–11:50. Clerk continued on paper; two entries were copied back afterwards, one incompletely.	Partial writes are the realistic failure mode, not total loss.	Ask the nurse what she found last time. Inspect the register for incomplete rows.

LOOK FOR THE NOTEBOOK

Rows 3 and 4 are starred because they are the same kind of finding: **a workaround**. Somebody has built a private system — a notebook, a second list, a habit of scanning for the mother — to survive a gap in the official one. A workaround is the strongest evidence a requirements engineer ever finds, because the person has already paid for it with their own time. It is not an opinion about what might be useful; it is a demonstrated cost.

And notice what row 4 does that row 3 does not: it carries **a number**. “19 of 31” is what moved S-05 above card printing in the backlog. An unquantified workaround is a story; a quantified one is a priority.

3 · Prototyping: provoking a reaction

Interviews and observation both have the same blind spot: neither can tell you how somebody will react to something that does not exist yet. For that you need to put an artifact in front of them — and the cheaper and uglier it is, the better the data.

DEFINITION — THROWAWAY PROTOTYPE

An artifact built to be **discarded**, whose only purpose is to provoke a reaction that reveals a requirement. It is not a first version of the system, and no code from it survives. Paper is usually the right medium, precisely because nobody mistakes it for a finished product.

The rule that makes it work: give the person a **task**, then say nothing. You are watching behaviour, not collecting opinions. “Do you like this screen?” produces politeness; “show me how you would find Awa’s record” produces a requirement.

APPLIED TO SANTE-BASSAM

Prototype A — the S-05 search screen, worked in full so you can copy the method

- 1 **Draw it in ten minutes, by hand.** One sheet: a search box, a list of five result rows, each row showing name, village, birth year and mother. No colours, no logo, no menu. If it took longer than ten minutes it is too finished to criticise.
- 2 **Give a task, then be silent.** “A woman has arrived with a two-year-old called Awa. She has no card. She is from Modeste. Find the child’s record.” Then do not speak, even when she pauses. Especially when she pauses.
- 3 **Record in three columns** as she works, exactly as in §2. One person operates the paper, one writes.
- 4 **Ask only afterwards**, and only about what you saw her do: “you started typing and then stopped — what were you thinking?”

#	Observation	Inference	How to test it
1	She typed awa , saw 14 results, and stopped.	First-name search alone is useless at this register size.	Count how many Awas exist in the real register.
2	She then looked for somewhere to type the mother’s name and found none.	The mother’s name must be a search field, not a displayed column.	Redraw with two fields; re-run the same task.
3	Asked which of two identical rows was right, she said “the one from Modeste” without hesitating.	Village disambiguates; birth year does not, because she does not know it.	Re-run with village removed from the row and see if she can still choose.
4	She read only the first three rows, then looked up at us.	Results beyond three are not read. Ranking matters more than completeness.	Re-run with the correct answer placed fifth.
5	Unprompted: “and if she is not here, I write it on a paper and give it to the nurse.”	There is a referral path we have not modelled at all.	Ask the nurse. Possibly a new story.

What one hour of paper bought: acceptance criterion AC1 of S-05 changed — village now sorts first, not birth year. One backlog hypothesis was refuted: nobody wants search by phone number, which two of you had assumed. And one new story appeared from row 5, which no interview would have produced, because the clerk does not think of the referral slip as part of “finding a patient”.

The three ways a prototype session fails

You talk. Every sentence you say while they work is a leading question with extra steps. **It is too pretty.** People will not criticise something that looks finished; they assume the decisions are already made. **You ask for opinions.** “What do you think?” gets you a view on the layout. A task gets you the workflow.

SESSION 8 TAKEAWAY

Never ask what you could have read; observe before you ask; and when you do ask, never put the answer in the question. Write three columns, always — what you saw, what you think it means, and how you will find out. Then look for the notebook: somebody has already built a private workaround for the gap in the system, and if you can attach a number to it, you have both a requirement and its priority.

4 • In-class exercise • the interview role-play

Two groups of three, twelve minutes each round, then five minutes reporting. Roles rotate, so everyone interviews once.

Roles

The clerk

reads a private brief from the instructor; three exceptions and one workaround the interviewer does not know about. Answers in character — helpfully, and never volunteering

The interviewer

has the field list from document analysis. May not ask anything a form would have answered

The observer

logs every question as

O

,

P

,

C

or

L

. Says nothing during the round

1. **Before starting**, the interviewer writes down what document analysis already told them.
2. **Twelve minutes** to uncover the normal workflow, two exceptions, what the clerk actually uses to find a patient, and what she does when the power fails — **without proposing a screen, a field or a feature in the first five minutes**.
3. **The observer reports** the O / P / C / L counts. For each L: what it invented, and how it would be repaired.
4. **All three** produce a four-row three-column record, and name the one item in the private brief that was missed, plus the question that would have found it.

EXPECT THIS

Most interviewers ask their first leading question inside ninety seconds — usually “so you would want...”.

Most miss the workaround entirely, because they never ask *what do you do when...*, and the clerk, in character, has no reason to mention it.

5 • Homework • individual • due before Session 9 (Tuesday 6 October)

Yours alone, about forty-five minutes. It continues the role-play and gives you the craft you will need in the field work your pair is committed to this fortnight.

TEMPLATE

Download **CSC4305_HW08_Template.docx** from **CSC_4305_SWE-Assignments**, fill it in, rename it **CSC4305_HW08_FirstName_LastName.docx** and commit it to **/homework/S08/** in your own private repository.
A .docx file. Not a PDF, not a link.

Q1 Repair five leading questions. For each, say what it invents or assumes, then rewrite it so that it discovers the same territory without supplying the answer.

(a) “You’d find it useful to have the report emailed automatically, wouldn’t you?” (b) “How much time does the duplicate problem waste?” (c) “Would you prefer a dropdown or a search box?” (d) “The current system is pretty frustrating, isn’t it?” (e) “Do you need offline access?”

Q2 Write a six-question protocol for *one* stakeholder your pair is not already assigned to — the district officer, the pharmacy assistant or the post supervisor. Type each question **O**, **P** or **C**. At least two must target an exception, and the last must be a referral question.

Six questions, not twelve. A protocol you could actually get through in twenty minutes is worth more than one you abandon.

Q3 Fix three broken records. Each of these mixes observation with inference. Split each into the three columns properly.

(a) “The clerk is frustrated by the slow register.” (b) “Patients don’t bring their cards because they lose them.” (c) “The nurse needs a better way to track follow-ups.”

Q4 Find one workaround in your own life. Any system you use — the university portal, a banking app, the library. Describe the workaround you or someone else has built around a gap in it, in three columns, and attach a number to it: how often, how long, how many people. Then write the user story it implies.

This is the one that teaches the instinct. Once you have noticed one, you cannot stop noticing them.

Q5 Design a prototype session in half a page. For story S-12, the nurse’s Monday follow-up list: what you draw, the exact task sentence you will give her, what you will be watching for, and the one thing that would make you conclude the story is wrong.

The last part is the hard one. If no possible outcome would change your mind, you are not running an experiment.

Q6 Reading. Sommerville, *Software Engineering* 10th ed., §4.4–4.5. Optional and short: Beyer & Holtzblatt, *Contextual Design*, chapter 3.

Where this goes next. Your pair’s records are the evidence base for Sessions 9 and 10. From Tuesday onward, every requirement must trace to a row in one of them — an observation, an interview question, a regulation clause or a prototype session. That is what “attributable” means, and a requirement without a source is a guess with an identifier.

Key terms

semi-structured interview

open question

probing question

confirming question

playback

leading question

referral question

observation vs inference

three-column record

workaround

throwaway prototype

contextual inquiry

Readings and references

- Sommerville, I. *Software Engineering*, 10th ed. — Chapter 4, §4.1–4.5.
- Beyer, H. & Holtzblatt, K. *Contextual Design*, 1998 — Chapter 3.
- Gottesdiener, E. *Requirements by Collaboration*, 2002.
- Snyder, C. *Paper Prototyping*, 2003 — running the session.